Sequencing the Future

An Overview of Next-Generation Sequencing (NGS) and Whole-Genome Sequencing (WGS)

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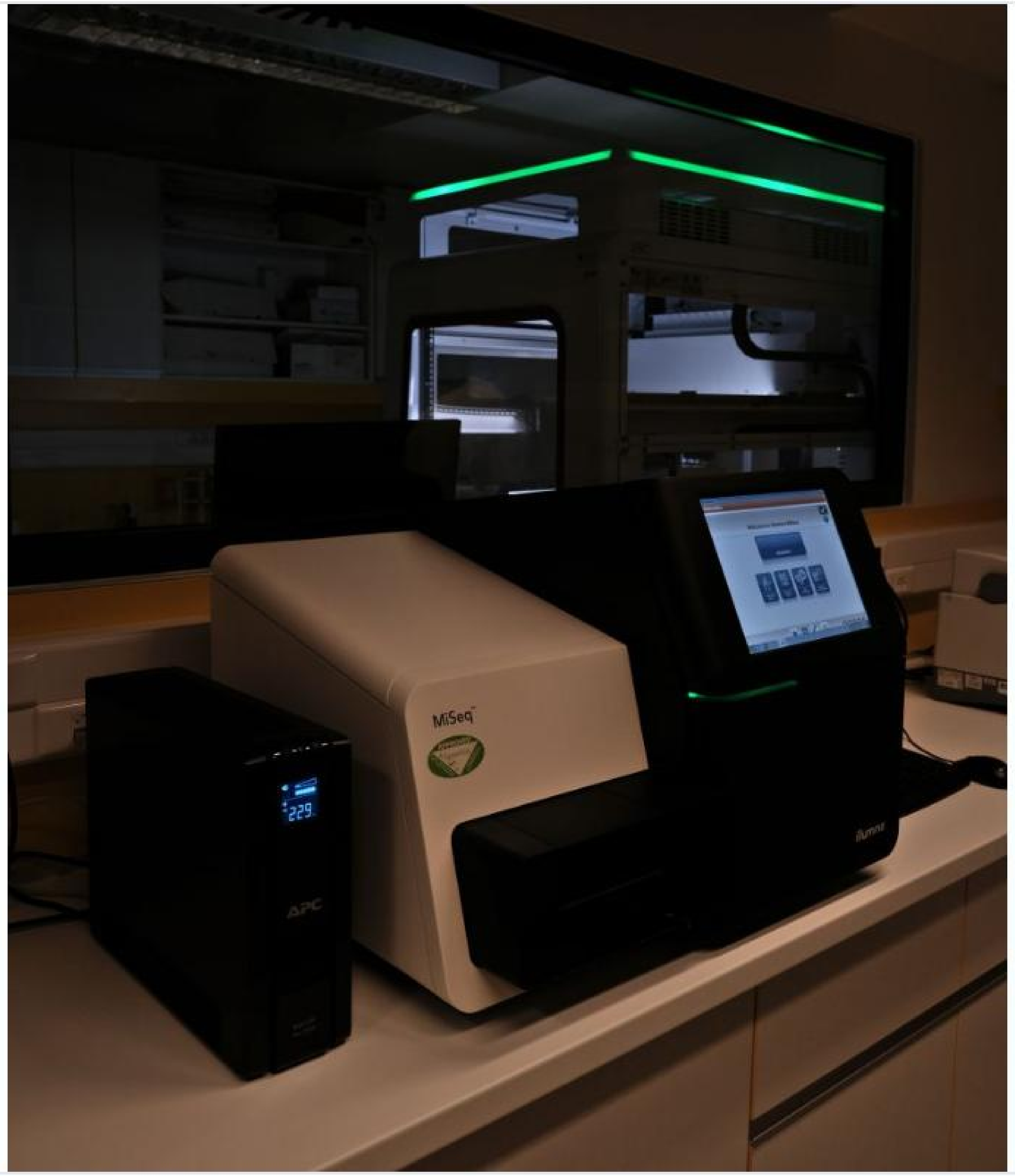
What is Next-Gen Sequencing (NGS)?

Next-Generation Sequencing (NGS) is a transformative technology that has revolutionized genomics research.

- Enables high-throughput, cost-effective analysis of DNA and RNA.
- Allows for sequencing millions (to billions) of DNA fragments simultaneously.
- Provides comprehensive insights into genome structure, genetic variations, and gene expression profiles.
- Has propelled advancements in rare diseases, cancer genomics, and microbiome analysis.



The General NGS Workflow

1. Library Preparation

Sample DNA or RNA is fragmented into smaller pieces. Special adapters are then ligated (attached) to the ends of each fragment.

2. Amplification

The adapter-ligated fragments are amplified to create millions of copies, forming clusters (e.g., Bridge PCR) or on beads (Emulsion PCR).

3. Sequencing

Platforms like Illumina use "Sequencing by Synthesis," where fluorescently labeled nucleotides are added one base at a time and imaged to read the sequence.

Key Advantages of NGS

- **High Throughput:** Ability to sequence millions to billions of DNA fragments in parallel in a single run.
- **Cost-Effective:** Dramatically reduced the cost per sequenced base compared to older methods like Sanger sequencing.
- **Comprehensive Insights:** Delivers deep insights into genomes (WGS), transcriptomes (RNA-Seq), and epigenomes (ChIP-Seq).
- **High Versatility:** A single platform can be used for diverse applications, from cancer genomics to infectious disease surveillance.

Diverse Applications of NGS

Research Applications

- ✓ **Genomics:** Whole-genome and exome sequencing.
- ✓ **Transcriptomics:** RNA-Seq for gene expression and alternative splicing analysis.
- ✓ **Epigenomics:** Studying DNA methylation and histone modifications.
- ✓ **Metagenomics:** Analyzing the microbiome (e.g., 16S rRNA).

Clinical & Diagnostic Uses

- ✓ **Infectious Disease:** Surveillance, pathogen detection, and drug resistance.
- ✓ **Inherited Diseases:** Diagnosis of rare genetic disorders.
- ✓ **Cancer:** Tumor profiling, prognosis, and identifying therapeutic targets.
- ✓ **Forensics:** DNA profiling and ancestry tracing.

Deep Dive: Whole-Genome Sequencing

Understanding the complete genetic blueprint

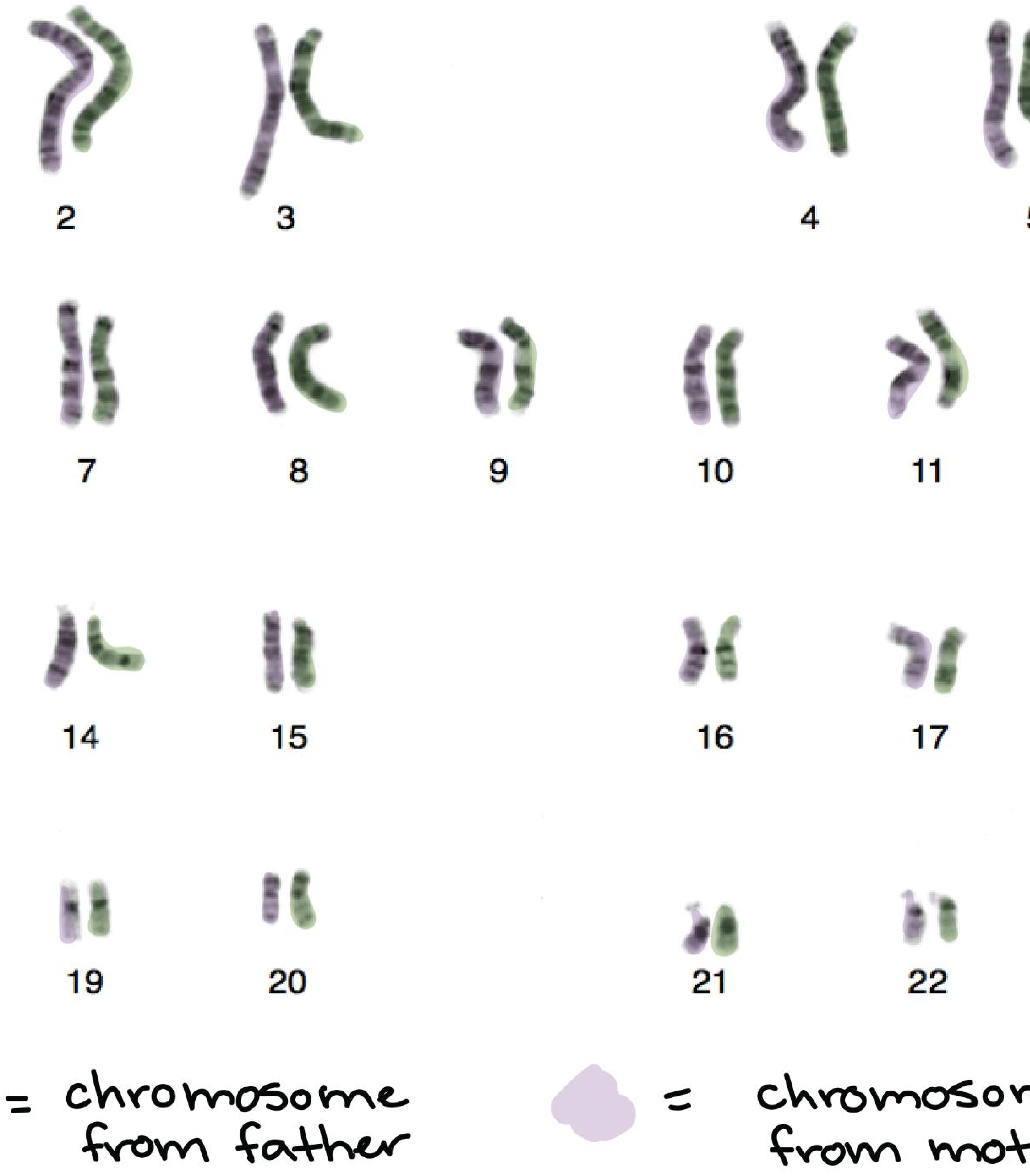
What is Whole-Genome Sequencing?

Whole-Genome Sequencing (WGS) is a comprehensive genomic technique that involves determining the **complete DNA sequence** of an individual's genome.

It provides a detailed blueprint of an organism's genetic makeup, encompassing:

- All protein-coding genes.
- Regulatory regions that control genes.
- Non-coding DNA elements.

WGS enables the identification of genetic variations, from single-nucleotide polymorphisms (SNPs) to large structural changes like deletions and rearrangements.



The WGS Workflow

1. DNA Extraction

High-quality, high-molecular-weight DNA is isolated from a sample (e.g., blood, tissue, or cell culture).

2. Library Prep

The DNA is fragmented, and NGS adapters are ligated to the ends, creating a sequencing library.

3. Sequencing

The library is sequenced using an NGS platform (e.g., Illumina) to generate millions of short or long reads.

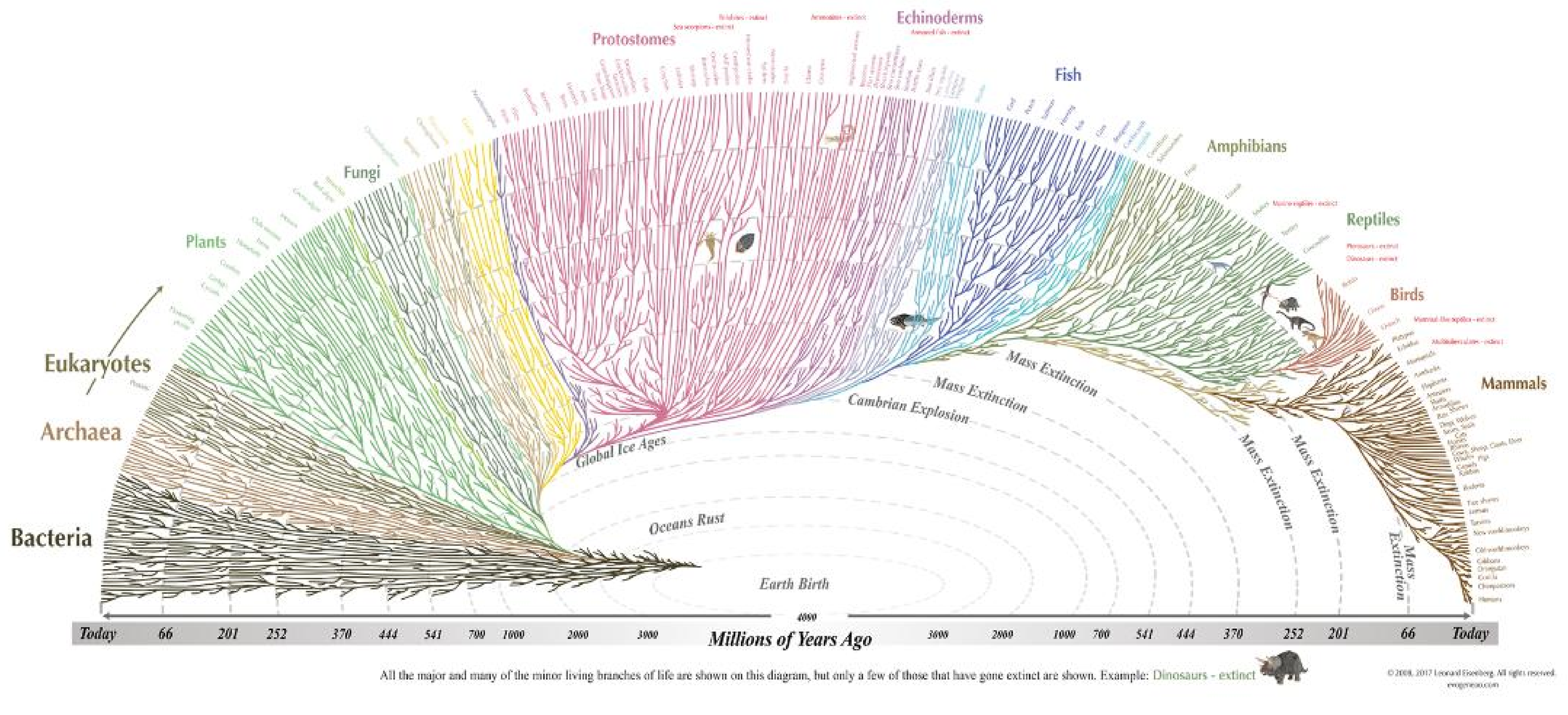
4. Data Analysis

Reads are aligned to a reference genome. Variant calling identifies SNPs, indels, and structural variants.

Applications of WGS

WGS is a powerful tool for comprehensive genetic analysis and discovery:

- **Discovery Science:** Assembling novel genomes (prokaryotes and eukaryotes) without a reference sequence.
- **Cancer Research:** Identifying all somatic (tumor-specific) and germline (inherited) mutations.
- **Rare Genetic Diseases:** Diagnosing patients with complex symptoms that are not explained by other tests.
- **Population Genetics:** Studying genetic diversity, ancestry, and evolutionary relationships.



Comparison: First-Gen vs. Next-Gen (NGS)

Feature	First-Gen (Sanger)	Next-Gen (e.g., Illumina)
Throughput	Low (One fragment per reaction)	Massively Parallel (Millions of fragments)
Read Length	Long (up to ~1000 bp)	Short (typically 36-300 bp)
Cost per Base	High	Extremely Low
Primary Use	Validating single genes, small projects	Genomes, Exomes, Transcriptomes

NGS vs. WGS: Method vs. Application

Next-Generation Seq. (NGS)

This is the **TECHNOLOGY** or **METHOD**.

- ✓ It's the "how" we read DNA.
- ✓ A broad term for modern, high-throughput sequencing (e.g., Illumina, PacBio).
- ✓ It is the "engine" that can be used for many different purposes.

Whole-Genome Seq. (WGS)

This is the **APPLICATION** or **STRATEGY**.

- ✓ It's "what" we are sequencing: the *entire* genome.
- ✓ It is one of many applications that *uses* NGS technology.
- ✓ Other NGS applications include WES (Exome), RNA-Seq (Transcriptome), etc.

Comparison: WGS vs. WES

Whole-Genome (WGS)

Sequences **~99%** of the genome.

- ✓ Covers all coding & non-coding DNA.
- ✓ Identifies structural variants & CNVs.
- ✓ **Best For:** Novel discovery, complex diseases, cancer genomics.
- ✓ **Data:** Very large, high cost.

Whole-Exome (WES)

Sequences **~1-2%** of the genome (the "exome").

- ✓ Covers protein-coding regions only.
- ✓ Misses non-coding regulatory regions.
- ✓ **Best For:** Identifying known disease-causing variants in genes.
- ✓ **Data:** Manageable, moderate cost.

Questions?

Thank you for your attention.

Image Sources

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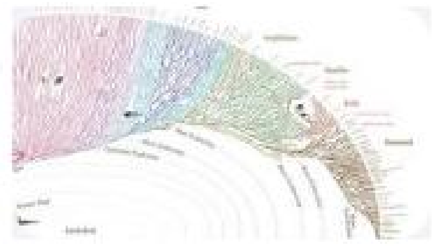

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